

Employment

Graduate Research Assistant	NC State University	Apr 2025 – Present
- Collaborated with plant biologists and geneticists to deploy cloud-native genomic data platform on GCP (Cloud Run) with JBrowse 2 genome browser, enabling 100+ researchers to visualize 500GB+ genome datasets. - Improved ML model accuracy by 12% via hyperparameter tuning and time-series feature analysis on biological sensor data.		
Product Engineer Intern	Isomer AI	May 2025 – May 2026
- Shipped per-tenant AI workflow platform for insurance carriers: Temporal.io durable orchestration, LLM risk signal detection across Claude/GPT-4o/Gemini, Auth0 RBAC, MCP servers; 30+ deployments. - Built MCP servers (Python + TypeScript); Playwright + Tailscale VPN for secure customer-internal browser automation.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Partnered with healthcare data teams to build 5+ AI workflows including health claim automation using OCR, NLP, and ML pipelines; improved document processing accuracy by 25% across 10K+ daily medical claims. - Automated AWS and GCP CI/CD pipelines; integrated Grafana monitoring and led model validation and red-teaming sessions.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led cross-functional SDLC for data-driven enterprise ticket analysis using NLP classification models; worked with support teams to reduce resolution time by 15% and eliminate 2,000+ monthly manual triages. - Built Kafka + Spark + Hadoop pipelines with data engineers; cut ticket-processing latency by 20% at peak daily load.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
Master of Science in Computer Science; GPA: 3.5/4.0. Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering, Programming Languages, Computer Networks, Distributed Systems.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infrastructure, DS & Algorithms, Machine Learning, Blockchain, Computer Networks, Database Systems.		

Projects

PulseRank	Real-Time Leaderboard & Activity Feed	2026
- Built event-driven leaderboard and activity feed in TypeScript with NestJS and Next.js; MySQL transactional writes, MongoDB feed projections with TTL, Redis sorted sets for sub-millisecond reads; live SSE. - Validated multi-DB consistency with Testcontainers; benchmarked hot paths with k6 at 10K concurrent users.		
AlgoPulse	Full-Stack Agentic AI Platform	2025
- Built agentic AI orchestration platform on Temporal.io and Kafka: agent loop with multi-step reasoning, tool invocation, and retry logic; sustained 2,000 async inference requests per second. - Deployed on AWS with auto-scaling groups; k6 load tested to 10,000 concurrent users; sub-100ms leaderboard latency. - Designed event-driven SSE delivery via Supabase Edge Functions; replaced polling and cut active server connections by 60%.		
KubeServe	Kubernetes LLM Inference Platform	2025
- Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing. - Deployed via Helm + ArgoCD GitOps on Kubernetes; OTel + Prometheus + Grafana observability stack; 88 pytest tests.		

Languages and Technologies

- Languages:** TypeScript, JavaScript, Python, Java, SQL, Bash
- Frontend & Runtime:** React, Next.js, NestJS, Node.js, Vite, Express, Auth0, Playwright, REST APIs
- Data & Storage:** PostgreSQL, MySQL, MongoDB, Redis, DuckDB, Kafka, Parquet, Testcontainers
- Infra & Observability:** Kubernetes, Docker, GCP, AWS, OpenTelemetry, Prometheus, Grafana, GitHub Actions

Additional Experience and Awards

- IEEE Published:** Co-authored and presented research on Computer Vision-Based Cybersecurity Threat Detection at the Springer icSoftComp 2023 international conference, reaching an audience of 500+ global attendees.
- Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; built 3 full-stack MVPs end-to-end in 48-hour competitive sprints.
- GDSC Lead:** Led 4+ tech events as Google Developer Student Club Lead, reaching 500+ students across hackathons and conferences.
- Technical Writing:** Published articles on LLM model collapse, AI agent security, and audio AI at kulkarniaditya.com/blog.